

Econometrics - Project	
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Project Title	Wage and Salary Income Among Immigrants in the United States, 2019-2023
Abstract (100 words max)	Immigrants constitute a significant portion of the U.S. labour force. This project analyses their earnings during 2019–2023, capturing the COVID-19 shock. Using IPUMS microdata and OLS regressions, we examine how arrival timing and integration affect hourly wages. We find that recent arrivals, particularly the pandemic cohort, face significant wage penalties, whereas citizenship and English proficiency are associated with higher earnings. These findings, confirmed by robustness checks on prime-age workers and language binary restrictions, highlight the critical role of assimilation time and human capital in wage determination.

Instructions

Please follow these formatting guidelines:

- Use Times New Roman, font size 11 pt, 1.15 line spacing.
- Use black text.
- Adhere to the specified word count.

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- All project content is considered preliminary and may be changed until the final submission deadline.
- Only the final submission will be graded.

For submission:

- For each deliverable, one representative of each group converts this document to a PDF and submit it by dragging and dropping the file on Virtuale in the appropriate link.
- For the first deliverable, you may name the file as you wish.
- After the first deliverable, you will be assigned a group number.
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 - o 1 raw dataset in dta format (e.g., **data_group_1.dta**). (Yes, dta is the Stata data format)
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Sample selection and descriptive statistics – 300 words max

Word count: 293

We utilize IPUMS USA microdata from the Multi-Year American Community Survey (ACS) for 2019–2023, with the individual respondent as the unit of analysis. All monetary variables are inflation-adjusted. We drop observations with missing values or zero wage, as logarithmic transformation cannot applied to zero.

The sample includes men and women aged 18–65 who report positive wage and salary income (*incwage* > 0), then the outcome variable is hourly wage and salary income (*wage_hour*).

Immigrant status is identified through the key variable *years_in_us*, defined as 2023 minus the year of immigration (*yrimmig*). A positive value indicates that the respondent is foreign-born and has resided in the United States for at least one year.

The first explanatory variable is the citizenship status, captured by a dummy variable (*citizen_dummy*) equal to 1 for U.S. citizens and 0 for non-citizens. In addition, we account for the timing of arrival in the U.S. through a categorical variable (*period_cat*) that groups immigrants into historical periods: the Great Depression (1929–1945), the Golden Age (1945–1973), Economic Stagnation (1973–1994), Globalization (1994–2007), the Great Recession (2007–2019), and the Pandemic (2020–2023). English knowledge is measured by *speakeng_cat*, which distinguishes four levels: does not speak English, speaks English poorly, speaks English well, and speaks only English. Both variables are converted into a set of dummy variables.

We include as controls: education (*educ_cat*), measuring the highest degree attained, *age* and *sex_dummy* (coded as 0 for males and 1 for females).

Table 1. Summary Statistics Table

Variable	Obs	Mean	Std. dev.	Min	Max
ln_wage_hour	1,016,525	3.275234	.907734	-6.659294	12.44378
years_in_us	1,016,525	23.83092	13.9757	0	70
period_cat					
2	1,016,525	.0494577	.2168218	0	1
3	1,016,525	.2930572	.4551647	0	1
4	1,016,525	.3464037	.4758239	0	1
5	1,016,525	.2738309	.4459235	0	1
6	1,016,525	.0372504	.189375	0	1
speakeng_cat					
1	1,016,525	.0289112	.1675572	0	1
2	1,016,525	.0973149	.2963862	0	1
3	1,016,525	.6460235	.4782022	0	1
4	1,016,525	.2277504	.4193809	0	1
educ_cat	1,016,525	2.838518	.799003	1	4
citizen_dummy	1,016,525	.6142594	.48677	0	1
sex_dummy	1,016,525	.4793271	.4995727	0	1
age	1,016,525	42.92485	12.22616	18	65

Table 1 reports a sample with 1,016,525 observations and a log-wage mean of 3.28. Individuals are gender-balanced (47.9% female, age 42.9) and integrated (23.8 years in US, 61.4% citizens). English proficiency is high, with 64.6% falling into category 3, while education averages 2.84.

We analyse how immigrants' hourly earnings vary with time spent in the United States and with observable characteristics related to labour market integration. Immigrants do not enter the host-country labour market on equal footing with native workers: at arrival, they may face disadvantages due to limited knowledge of the local labour market, language barriers, and restricted access to certain jobs. Over time, as immigrants accumulate experience in the host country and improve their integration, these disadvantages may decrease, leading earnings to converge toward native-born wages. As a result, wages can differ systematically across immigrants depending on their timing of arrival and degree of integration. More recent arrivals may earn less than earlier arrivals, while education, English proficiency, and citizenship are expected to be associated with better-paid job opportunities (Borjas, 2016).

Identification relies on differences in hourly wages across immigrants observed in the same period who arrived in the United States at different times and display different levels of labour market integration. Conditional on age, sex, and other observed characteristics, we assume that these wage differences are not systematically driven by unobserved determinants of earnings. Under this assumption, OLS estimates capture conditional correlations rather than causal effects. Focusing on the 2019-2023 period, which spans the COVID-19 shock and the subsequent recovery, enhances the relevance of these cross-sectional comparisons, as wage differences across immigrant groups by arrival cohort and labour market position are likely to be more pronounced. The dependent variable is the natural logarithm of hourly wage and salary income, $\ln(wage_hour)$.

First, we checked for homoskedasticity (Breusch-Pagan test) and found evidence of heteroskedasticity, meaning that the variance of the error term is not constant across observations. To account for this issue, we report heteroskedasticity-robust standard errors in all regressions.

Second, we examined multicollinearity among the explanatory variables (VIF test). We found a high degree of multicollinearity between the categorical variable capturing the period of arrival and the variable measuring years spent in the United States. Since these variables convey overlapping information, we decided to drop *years_in_us* from the specification and retain the period of arrival dummies in the final model. Now we estimate the following log-linear model using OLS with *citizen_dummy* as the main key variable:

$$\ln(wage_hour)_i = \beta_0 + \beta_1 citizen_dummy_i + \sum_{j=3}^6 \delta_j D_{period,j} + \sum_{k=2}^4 \gamma_k D_{speakeng,k} + \beta_2 educ_cat_i + \beta_3 sex_dummy_i + \beta_4 age_i + \epsilon_i.$$

In this specification, coefficients can be interpreted as approximate percentage changes in hourly wages associated with changes in the explanatory variables, holding all other factors constant.

We test two joint hypotheses. The first hypothesis examines whether the timing of arrival, citizenship status, and English proficiency jointly affect hourly wages. Formally:

$$H_1: \beta_{citizen_dummy} = \sum_{j=3}^6 \delta_j D_{period,j} = \sum_{k=2}^4 \gamma_k D_{speakeng,k} = 0.$$

The second hypothesis tests whether age, sex, and education jointly affect wages:

$$H_2: \beta_{educ_dummy} = \beta_{sex_dummy} = \beta_{age} = 0.$$

A main limitation of our analysis is that the data are repeated cross-sections, so we do not follow the same individuals over time and cannot directly observe individual earnings trajectories or migration decisions. As a result, cross-sectional comparisons across arrival cohorts may be affected by selection mechanisms. For instance, Akee and Jones (2019) show that immigrants who eventually leave the United States experience declining earnings prior to return, indicating negative selection among return migrants. Similarly, Rho and Sanders (2021) document that labour market participation is not random, as some immigrants, particularly among the highly educated, enter

employment with a delay after arrival. In addition, unobserved factors such as ability, occupation, or local labour market conditions may be correlated with integration measures. These mechanisms can bias cross-sectional estimates and lead to an understatement of true earnings assimilation. We address these concerns by controlling for key observable characteristics, while acknowledging that some selection and unobserved factors may still affect the results.

Results and robustness checks – 600 words max

Word count: 597

The F-tests show that the groups of variables included in the model are jointly significant in explaining immigrants’ hourly wages. Both joint null hypotheses are rejected at the 5% level, indicating that integration-related factors (arrival timing, citizenship, English proficiency) and standard demographic characteristics (age, sex, education) are jointly relevant in explaining wages. These results support the chosen specification and motivate the interpretation of the regression coefficients in the next section.

Table 2. Regression Table

	(1) ln_wage_hour b/se
period_cat=2	0.000 (.)
period_cat=3	0.048*** (0.004)
period_cat=4	0.022*** (0.004)
period_cat=5	-0.020*** (0.005)
period_cat=6	-0.156*** (0.007)
speakeng_cat=1	0.000 (.)
speakeng_cat=2	-0.009* (0.005)
speakeng_cat=3	0.183*** (0.005)
speakeng_cat=4	0.241*** (0.005)
citizen_dummy	0.037*** (0.002)
educ_cat	0.403*** (0.001)
sex_dummy	-0.245*** (0.002)
age	0.010*** (0.000)
Constant	1.614*** (0.008)
Observations	1016525
R-squared	0.191
Adjusted R-squared	0.191

Table 2 reports the results of the baseline OLS regression, which show a clear relationship between arrival timing and hourly wages. Relative to earlier arrivals, immigrants who entered the United States in intermediate periods (*period_cat* = 3 and 4) earn slightly higher hourly wages, though the effects are modest. In contrast, more recent arrivals (*period_cat* = 6) earn significantly lower wages, with the largest and most precisely estimated penalty observed for immigrants arriving in 2020–2023. This finding is consistent with the immigrant earnings assimilation framework, which predicts initial disadvantages due to limited host-country experience, weaker professional networks, and imperfect knowledge of the local labour market (Borjas 1995; Borjas, 2016). The particularly

strong penalty for the pandemic cohort likely reflects adverse entry conditions, including disrupted job matching and concentration in lower-wage sectors during the COVID-19 shock.

English knowledge is strongly and positively associated with hourly wages. Compared to immigrants who do not speak English, those who speak English well or exclusively earn significantly more, with coefficients that are both statistically and economically meaningful. This result is in line with extensive evidence showing that language skills improve productivity and access to better-paid jobs (Chiswick and Miller, 2007; Bleakley and Chin, 2004).

Citizenship status is also positively related to earnings. The coefficient on the citizenship dummy is positive and statistically significant, implying that naturalized immigrants earn higher wages than non-citizens, conditional on observable characteristics. This finding is consistent with previous research showing that citizenship improves labour market outcomes by expanding job opportunities and reducing legal uncertainty (Bratsberg, Ragan, and Nasir, 2002). However, citizenship should not be interpreted as exogenous, as it may also reflect prior integration and unobserved individual traits.

Among the control variables, education has a large positive effect on wages, consistent with human capital theory. Age is positively associated with earnings, reflecting experience accumulation. The coefficient on sex is negative and statistically significant, indicating a persistent gender wage gap among immigrants, even after controlling for other characteristics (Dustmann and Fabbri, 2003).

Table 3. Robustness Check Table

	(1) ORIGINAL b/se	(2) RESTRICTED b/se	(3) DUMMY_S~N b/se
period_cat=2	0.000 (.)	0.000 (.)	0.000 (.)
period_cat=3	0.048*** (0.004)	0.034*** (0.007)	0.013*** (0.004)
period_cat=4	0.022*** (0.004)	0.033*** (0.007)	-0.028*** (0.004)
period_cat=5	-0.020*** (0.005)	-0.002 (0.007)	-0.079*** (0.004)
period_cat=6	-0.156*** (0.007)	-0.148*** (0.009)	-0.218*** (0.007)
speakeng_cat=1	0.000 (.)	0.000 (.)	
speakeng_cat=2	-0.009* (0.005)	-0.006 (0.006)	
speakeng_cat=3	0.183*** (0.005)	0.197*** (0.006)	
speakeng_cat=4	0.241*** (0.005)	0.261*** (0.006)	
citizen_dummy	0.037*** (0.002)	0.047*** (0.002)	0.058*** (0.002)
educ_cat	0.403*** (0.001)	0.402*** (0.001)	0.423*** (0.001)
sex_dummy	-0.245*** (0.002)	-0.255*** (0.002)	-0.247*** (0.002)
age	0.010*** (0.000)	0.011*** (0.000)	0.009*** (0.000)
speakeng_dummy			0.145*** (0.005)
Constant	1.614*** (0.008)	1.608*** (0.011)	1.669*** (0.008)
Observations	1016525	722115	1016525
R-squared	0.191	0.193	0.187
Adjusted R-squ~d	0.191	0.193	0.187

Table 3 reports the main results about two robustness checks that confirm the stability of the baseline results. First, we restrict the sample to prime-age workers (25–54) (Column 2). This restriction is standard in labour economics, as labour market participation and wage determination are more stable within this age range. Younger individuals may still be in education or working part-time, while older workers may reduce labour supply or exit the labour market. This yields coefficients similar in sign and magnitude to the baseline estimates, suggesting that the findings are not driven

by age-related participation patterns. Second, replacing the categorical English measure with a binary indicator for speaking English well produces similar results, indicating that our estimates are not sensitive to alternative language definitions (Column 3).

Several limitations remain. Unobserved factors such as occupation, industry, or location may bias the estimates if correlated with integration measures. Finally, the results should be interpreted as conditional correlations rather than causal effects. Despite these limitations, the consistency of the estimates across specifications and their alignment with the existing literature suggest that the results capture meaningful patterns in immigrant wage determination.

References – max 10

Please provide the list of references cited in your project

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